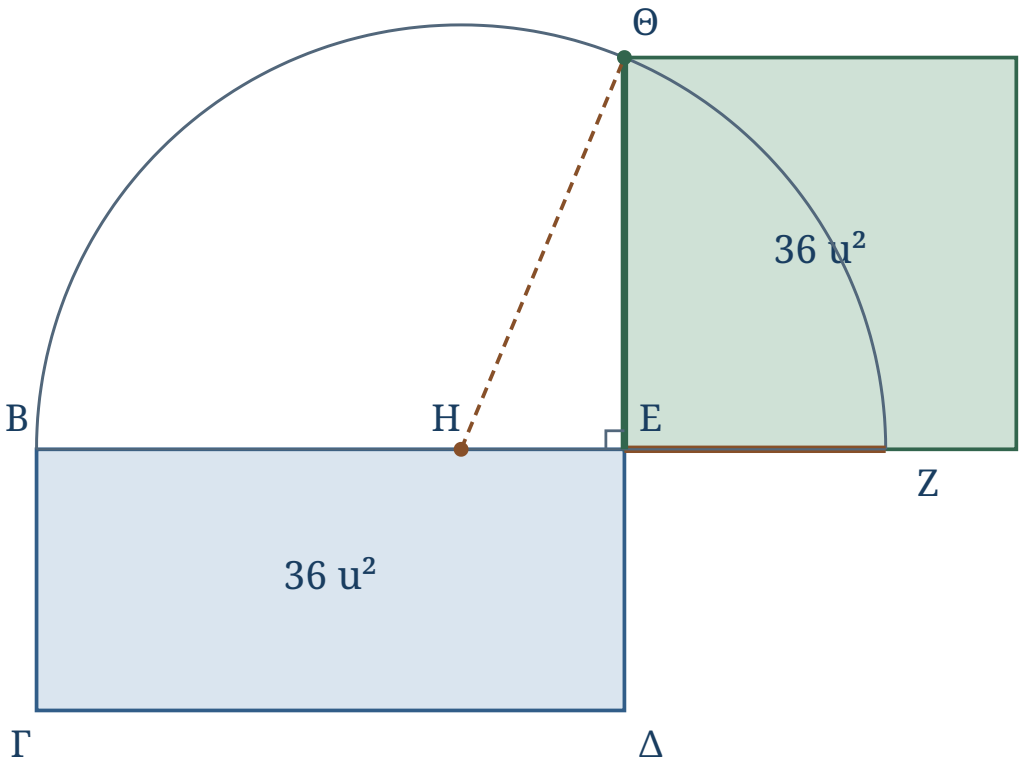
# A square, with nothing added.

Given rectangle: 9 by 4 drawing units. Area: 36 square units. Constructed side: approximately 6 units. The half-sum 6.5 makes its square 6.25 square units too large.

Open these settings in the workshop · Save the exact measurement table




BE = 9 u · EΔ = 4 u

u = an arbitrary drawing unit · equal scales

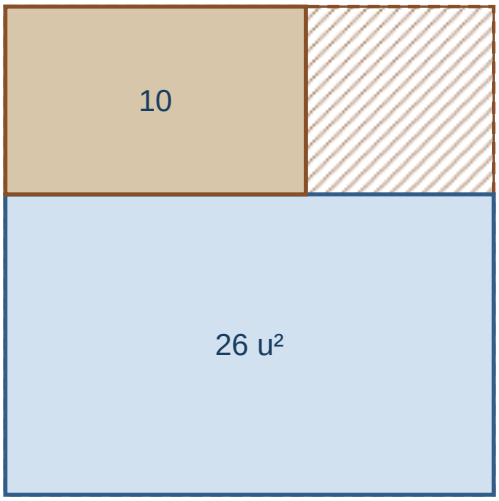
Selected stage: Account for the area. Equal horizontal and vertical scales; arbitrary drawing units. H is the centre; Θ is the circle intersection. Greek point labels follow Heiberg.

**Equal area and equal perimeter are different conditions**

| Figure | Side(s) | Area | Perimeter |
|---|---|---|---|
| **Given rectangle** | 9 × 4 u | 36 u² | 26 u |
| **Square on EΘ** | 6 u | 36 u² | 24 u |
| **Square on the half-sum** | 6.5 u | 42.25 u² | 26 u |

# The extra area in the average

Dashed square: side 6.5 u · hatched corner: 6.25 u²




Same two pieces, rearranged. The corner remains empty.

The completed modern two-piece rearrangement. Same area 36 u²; the 6.25 u² corner remains empty. This illustrates the II.5 relation, not its original diagram or full proof.

The two pieces have areas 26 and 10 square units. A square with the rectangle's half-sum as its side has the same perimeter, but is larger in area unless the sides are equal.

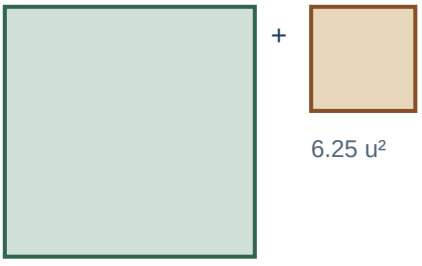
# Subtract the common square

II.5 · Two cuts of the line




BE × EZ = 36 u²

I.47 · Right triangle




EΘ² = 36 u²

Both totals: $H\Theta^2 = 42.25\ u^2$

Before subtraction: two accounts of 42.25 square units. The figures use the same scale.

II.5: rectangle + square on HE = square on HZ. Equal radii make the latter the square on HΘ. I.47: square on EΘ + square on HE = square on HΘ. Subtract the common 6.25 u² square. The remaining rectangle and constructed square are equal in area.

The rectangle's right angle makes HEΘ a right triangle. I.46 supplies the final square on the constructed side.

# The complete argument

Greek: Heiberg's 1883 edition. Latin: his nineteenth-century translation, not an ancient Latin text. English and reading divisions are new. The complete proposition includes a preliminary conversion of a general rectilinear figure by I.45. This workshop implements its rectangle case.

### 1. A square equal to a figure

Τῷ δοθέντι εὐθυγράμμῳ ἴσον τετράγωνον συστήσασθαι. Ἔστω τὸ δοθὲν εὐθύγραμμον τὸ Α· δεῖ δὴ τῷ Α εὐθυγράμμῳ ἴσον τετράγωνον συστήσασθαι.

Quadratum datae figurae rectilineae aequale construere. Sit data figura rectilinea A. oportet igitur figurae rectilineae A aequale quadratum construere.

To construct a square equal to the given rectilinear figure. Let A be the given rectilinear figure. We must therefore construct a square equal to the rectilinear figure A.

**ἴσον · aequale** — equal — here, in area. ἴσον agrees with τετράγωνον, ‘square’. The dative τῷ … εὐθυγράμμῳ names what it is equal to. Equal area does not require equal sides or the same shape. A names a whole figure, not a point. Our workshop starts with a rectangle; the proposition's opening claim is more general.

Printed pages 160, 161.

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### 2. An earlier construction supplies a rectangle

Συνεστάτω γὰρ τῷ Α εὐθυγράμμῳ ἴσον παραλληλόγραμμον ὀρθογώνιον τὸ ΒΔ·

construatur enim figurae rectilineae A aequale parallelogrammum rectangulum ΒΔ.

For let the rectangular parallelogram ΒΔ be constructed equal to the rectilinear figure A.

**ὀρθογώνιον · rectangulum** — right-angled. A parallelogram alone need not be a rectangle: ὀρθογώνιον adds the right-angle condition. Heiberg's Latin cites I.45 here. That earlier construction converts a rectilinear figure to an equal-area parallelogram at a specified angle. It is a prerequisite, not an operation carried out by this rectangle workshop. ΒΔ names the rectangle by opposite corners.

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3. Already a square? Stop here

εἰ μὲν οὖν ἴση ἐστὶν ἡ ΒΕ τῇ ΕΔ, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν. συνέσταται γὰρ τῷ Α εὐθυγράμμῳ ἴσον τετράγωνον τὸ ΒΔ· εἰ δὲ οὔ, μία τῶν ΒΕ, ΕΔ μείζων ἐστίν.

si igitur BE = EΔ, effectum erit, quod propositum erat. constructum enim est quadratum BΔ datae figurae rectilineae A aequale. sin minus, alterutra rectarum BE, EΔ maior est.

If BE equals EΔ, what was required would already be done: the square BΔ has been constructed equal to the rectilinear figure A. If not, one of BE and EΔ is greater.

**εἰ μὲν … εἰ δὲ οὔ** — if … but if not. The two conditions divide the argument. γεγονὸς ἂν εἴη uses a perfect participle with a potential optative: the task would stand accomplished. The semicircle is unnecessary when the given rectangle is a square. Choosing the larger side as BE fixes the orientation of the remaining construction.

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4. Copy a side and bisect the whole

ἔστω μείζων ἡ ΒΕ, καὶ ἐκβεβλήσθω ἐπὶ τὸ Ζ, καὶ κείσθω τῇ ΕΔ ἴση ἡ ΕΖ, καὶ τετμήσθω ἡ ΒΖ δίχα κατὰ τὸ Η,

sit maior BE, et producatur ad Z, et ponatur EZ = EΔ, et BZ in H in duas partes aequales secetur,

Let BE be greater. Extend it to Z, set EZ equal to EΔ, and bisect BZ at H.

**δίχα · in duas partes aequales** — into two equal parts. The single adverb δίχα is expanded in the Latin. We bisect the whole BZ, not the original side BE. E and H are distinct cuts of the same line when the sides differ. H is Greek eta: it is the centre label in this edition, not the English label H for the top of the semicircle.

Printed pages 160, 161.

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## 5. Draw the semicircle and meet it

καὶ κέντρῳ τῷ Η, διαστήματι δὲ ἑνὶ τῶν ΗΒ, ΗΖ ἡμικύκλιον γεγράφθω τὸ ΒΘΖ, καὶ ἐκβεβλήσθω ἡ ΔΕ ἐπὶ τὸ Θ, καὶ ἐπεζεύχθω ἡ ΗΘ.

et centro H radio autem alterutra rectarum HB, HZ semicirculus describatur BΘZ, et producatur ΔE ad Θ, et ducatur HΘ.

With centre H and radius either HB or HZ, draw the semicircle BΘZ. Extend ΔE to Θ, and join HΘ.

**κέντρῳ … διαστήματι** — with centre … with radius. Both datives specify how to draw. ἑνὶ τῶν means ‘one of the’, because the two half-lengths are equal. ΔE continues through E into the semicircle. The rectangle supplies a right angle at E. The lengths and coordinate orientation shown here are modern choices.

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## 6. The first account of one square

Ἐπεὶ οὖν εὐθεῖα ἡ ΒΖ τέτμηται εἰς μὲν ἴσα κατὰ τὸ Η, εἰς δὲ ἄνισα κατὰ τὸ Ε, τὸ ἄρα ὑπὸ τῶν ΒΕ, ΕΖ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΕΗ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΗΖ τετραγώνῳ.

iam quoniam recta BZ in partes aequales secta est in H in inaequales autem in E, erunt $BE \times EZ + EH^2 = HZ^2$.

Since BZ has been cut into equal parts at H and unequal parts at E, the rectangle contained by BE and EZ, together with the square on EH, equals the square on HZ.

**μετὰ τοῦ … τετραγώνου** — together with the square. μετά with the genitive adds a figure to the comparison. This is II.5, whose statement is available among the source pages. Heiberg compresses the Greek sentence into an equation in his Latin translation. The multiplication sign and squared notation belong to that nineteenth-century rendering; they are not printed in this Greek sentence.

Printed pages 160, 162, 161, 163.

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## 7. Equal radii exchange the square

ἴση δὲ ἡ ΗΖ τῇ ΗΘ· τὸ ἄρα ὑπὸ τῶν ΒΕ, ΕΖ μετὰ τοῦ ἀπὸ τῆς ΗΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΗΘ.

sed HZ = HΘ. itaque BE × EZ + HE² = HΘ².

But HZ equals HΘ. So the rectangle contained by BE and EZ, together with the square on HE, equals the square on HΘ.

**τὸ ἀπὸ τῆς ΗΘ** — the [square] on HΘ. The noun ‘square’ is understood after its earlier use. Both HZ and HΘ run from the centre to the circle. Reversing EH to HE changes the order of the endpoints, not the segment or its square.

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## 8. The second account of the same square

τῷ δὲ ἀπὸ τῆς ΗΘ ἴσα ἐστὶ τὰ ἀπὸ τῶν ΘΕ, ΕΗ τετράγωνα· τὸ ἄρα ὑπὸ τῶν ΒΕ, ΕΖ μετὰ τοῦ ἀπὸ ΗΕ ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΘΕ, ΕΗ.

uerum ΘE² + EH² = HΘ². itaque BE × EZ + HE² = ΘE² + EH².

The squares on ΘE and EH equal the square on HΘ. Therefore the rectangle contained by BE and EZ, together with the square on HE, equals the squares on ΘE and EH.

**τὰ … τετράγωνα** — the squares, taken together. The plural names two squares whose combined area is compared with one. I.47 applies to the right triangle HEΘ. It does not say that each of the smaller squares separately equals the square on the hypotenuse.

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9. Remove what both sides have

κοινὸν ἀφῃρήσθω τὸ ἀπὸ τῆς ΗΕ τετράγωνον· λοιπὸν ἄρα τὸ ὑπὸ τῶν ΒΕ, ΕΖ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ τετραγώνῳ.

subtrahatur, quod commune est, $HE^2$. itaque $BE \times EZ = E\Theta^2$.

Let the common square on HE be subtracted. The remaining rectangle contained by BE and EZ therefore equals the square on EΘ.

**κοινὸν ἀφῃρήσθω** — let the common part be subtracted. ἀφῃρήσθω is a third-person perfect passive imperative of ἀφαιρέω, ‘take away’. Latin subtrahatur is a passive subjunctive used as a command. The same area leaves both totals. This is an application of the common notion about subtracting equals from equals.

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10. Return to the given figure

ἀλλὰ τὸ ὑπὸ τῶν ΒΕ, ΕΖ τὸ ΒΔ ἐστιν· ἴση γὰρ ἡ ΕΖ τῇ ΕΔ· τὸ ἄρα ΒΔ παραλληλόγραμμον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΘΕ τετραγώνῳ. ἴσον δὲ τὸ ΒΔ τῷ Α εὐθυγράμμῳ. καὶ τὸ Α ἄρα εὐθύγραμμον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ ἀναγραφησομένῳ τετραγώνῳ.

uerum $BE \times EZ = B\Delta$; nam $EZ = E\Delta$. itaque $B\Delta = \Theta E^2$. sed $B\Delta = A$. itaque etiam figura rectilinea A quadrato, quod in EΘ construi poterit, aequale est.

But the rectangle contained by BE and EZ is BΔ, because EZ equals EΔ. Thus BΔ equals the square on ΘE. And BΔ equals the rectilinear figure A. So A also equals the square that can be drawn on EΘ.

**ἀναγραφησομένῳ** — that will be drawn. This future passive participle agrees with τετραγώνῳ. The argument has found the side of the required square; actually drawing the square uses the earlier construction I.46. Latin ‘quod … construi poterit’ expresses what can be constructed. The final square on our plate makes this last operation visible.

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11. The task has been accomplished

Τῷ ἄρα δοθέντι εὐθυγράμμῳ τῷ Α ἴσον τετράγωνον συνέσταται τὸ ἀπὸ τῆς ΕΘ ἀναγραφησόμενον· ὅπερ ἔδει ποιῆσαι.

Ergo datae figurae rectilineae A aequale quadratum constructum est, id quod in ΕΘ describi poterit; quod oportebat fieri.

A square equal to the given rectilinear figure A has therefore been constructed: the one that can be drawn on ΕΘ. This is what had to be done.

**ὅπερ ἔδει ποιῆσαι** — which is what had to be done. ποιῆσαι means ‘to do’ or ‘to make’. A construction ends with something made. A theorem commonly ends with δεῖξαι, ‘to show’. The opening infinitive συστήσασθαι sets the task; the perfect συνέσταται now presents the achieved result.

Printed pages 162, 163.

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# Source and model boundaries

J. L. Heiberg, *Euclidis opera omnia*, volume I, Leipzig: Teubner, 1883, pp. 160–163. Full pages 128–129 and 110–111 supply the statements of II.5 and I.47; their complete proofs are not transcribed. University of Toronto scan; public domain.

Line breaks, apparatus and inline Latin reference marks are omitted in the transcription. Greek ϑ becomes θ. Greek point labels remain Greek. Heiberg's Latin formulas, including his printed ‘aequale’ after ‘figura’ on p. 163, are retained. The full scans preserve the printed evidence.

For modern calculation, $r = (a + b)/2$, $d = (a − b)/2$ and $h^2 = r^2 − d^2$. Thus $h^2 = ab$. Chosen units, numerical approximations, algebraic vocabulary and the piece motion are teaching aids. The moving diagram is not a formal proof, an ancient measurement, a full implementation of I.45, or a construction for squaring a circle.

This file embeds its source scans and font, has no runtime scripts and makes no remote subresource requests. Its application link requires a connection. The completed rearrangement and the pre-subtraction account are retained for comparison, regardless of the captured view.

Ἐν ἄρα τοῖς ὀξυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀξεῖαν γωνίαν ὑποτεινούσης πλευρᾶς τετράγωνον ἔλαττόν ἐστι τῶν ἀπὸ τῶν τὴν ὀξεῖαν γωνίαν περιεχουσῶν πλευρῶν τετραγώνων τῷ περιεχομένῳ δὶς ὑπό τε μιᾶς τῶν περὶ τὴν ὀξεῖαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐντὸς ὑπὸ τῆς καθέτου πρὸς τῇ ὀξείᾳ γωνίᾳ· ὅπερ ἔδει δεῖξαι.

## ιδ'.

Τῷ δοθέντι εὐθυγράμμῳ ἴσον τετράγωνον συστήσασθαι.

Ἔστω τὸ δοθὲν εὐθύγραμμον τὸ $A$· δεῖ δὴ τῷ $A$ εὐθυγράμμῳ ἴσον τετράγωνον συστήσασθαι.

Συνεστάτω γὰρ τῷ $A$ εὐθυγράμμῳ ἴσον παραλληλόγραμμον ὀρθογώνιον τὸ $B\Delta$· εἰ μὲν οὖν ἴση ἐστὶν ἡ $BE$ τῇ $E\Delta$, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν. συνέσταται γὰρ τῷ $A$ εὐθυγράμμῳ ἴσον τετράγωνον τὸ $B\Delta$· εἰ δὲ οὔ, μία τῶν $BE$, $E\Delta$ μείζων ἐστίν. ἔστω μείζων ἡ $BE$, καὶ ἐκβεβλήσθω ἐπὶ τὸ $Z$, καὶ κείσθω τῇ $E\Delta$ ἴση ἡ $EZ$, καὶ τετμήσθω ἡ $BZ$ δίχα κατὰ τὸ $H$, καὶ κέντρῳ τῷ $H$, διαστήματι δὲ ἑνὶ τῶν $HB$, $HZ$ ἡμικύκλιον γεγράφθω τὸ $B\Theta Z$, καὶ ἐκβεβλήσθω ἡ $\Delta E$ ἐπὶ τὸ $\Theta$, καὶ ἐπεζεύχθω ἡ $H\Theta$.

Ἐπεὶ οὖν εὐθεῖα ἡ $BZ$ τέτμηται εἰς μὲν ἴσα κατὰ

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1. ἐν] inter ε et ν ras. 1 litt. V. 2. ἔλασσον F. 3. ἐστίν V. 4. τε] om. F. 6. ἐντός] om. P. 11. τὸ μὲν δοθέν p. 13. γάρ] om. p. 14. $B\Delta$] $B\Gamma\Delta E$ p; in ras. V. 15. συνέσταται] PBF, V m. 2; συνεστάτω V m. 1; συνίσταται p. 17. οὖ] postea add. F. Post μία 1 litt. (ι?) eras. F. 18. ἐκβεβλήσθαι φ. 19. EZ] ZE BF. 20. καί] postea add. F. κέντρῳ] PB, F m. 1; κέντρῳ μέν Vp, F m. 2. $HB$] $BH$ BF. 23. οὖν] om. F. Seq. ras. 1 litt. V. $BZ$] in ras. V. εἰς] -ς supra m. 1 V.

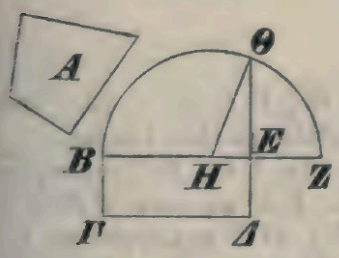
Ergo in triangulis acutiangulis quadratum lateris sub acuto angulo subtendentis quadratis laterum acutum angulum comprehendentium minus est duplo rectangulo comprehenso ab altero laterum acutum angulum comprehendentium, eo scilicet, in quod perpendicularis cadit, et recta a perpendiculari ad angulum acutum intra abscisa; quod erat demonstrandum.

## XIV.

Quadratum datae figurae rectilineae aequale construere.

Sit data figura rectilinea *A*. oportet igitur figurae rectilineae *A* aequale quadratum construere.

construatur enim figurae rectilineae *A* aequale parallelogrammum rectangulum $B\Delta$ [I, 45]. si igitur $BE = E\Delta$, effectum erit, quod propositum erat. constructum enim est quadratum $B\Delta$ datae figurae rectilineae *A* aequale. sin minus, alterutra rectarum $BE$, $E\Delta$ maior est. sit maior $BE$, et producatur ad $Z$, et ponatur $EZ = E\Delta$, et $BZ$ in $H$ in duas partes aequales secetur [I, 10], et centro $H$ radio autem alterutra rectarum $HB$, $HZ$ semicirculus describatur $B\Theta Z$, et producatur $\Delta E$ ad $\Theta$, et ducatur $H\Theta$.




iam quoniam recta $BZ$ in partes aequales secta

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XIV. Simplic. in Arist. de coel. fol. 101; id. in phys. fol. 12$^{a}$; 14. Boetius p. 386, 23.

τὸ *H*, εἰς δὲ ἄνισα κατὰ τὸ *E*, τὸ ἄρα ὑπὸ τῶν *BE*, *EZ* περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς *EH* τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς *HZ* τετραγώνῳ. ἴση δὲ ἡ *HZ* τῇ *HΘ*· τὸ ἄρα ὑπὸ τῶν *BE*, *EZ* μετὰ τοῦ ἀπὸ τῆς *HE* ἴσον ἐστὶ τῷ ἀπὸ τῆς *HΘ*. τῷ δὲ ἀπὸ τῆς *HΘ* ἴσα ἐστὶ τὰ ἀπὸ τῶν *ΘE*, *EH* τετράγωνα· τὸ ἄρα ὑπὸ τῶν *BE*, *EZ* μετὰ τοῦ ἀπὸ *HE* ἴσα ἐστὶ τοῖς ἀπὸ τῶν *ΘE*, *EH*. κοινὸν ἀφῃρήσθω τὸ ἀπὸ τῆς *HE* τετράγωνον· λοιπὸν ἄρα τὸ ὑπὸ τῶν *BE*, *EZ* περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς *EΘ* τετραγώνῳ. ἀλλὰ τὸ ὑπὸ τῶν *BE*, *EZ* τὸ *BΔ* ἐστιν· ἴση γὰρ ἡ *EZ* τῇ *EΔ*· τὸ ἄρα *BΔ* παραλληλόγραμμον ἴσον ἐστὶ τῷ ἀπὸ τῆς *ΘE* τετραγώνῳ. ἴσον δὲ τὸ *BΔ* τῷ *A* εὐθυγράμμῳ. καὶ τὸ *A* ἄρα εὐθύγραμμον ἴσον ἐστὶ τῷ ἀπὸ τῆς *EΘ* ἀναγραφησομένῳ τετραγώνῳ.

Τῷ ἄρα δοθέντι εὐθυγράμμῳ τῷ *A* ἴσον τετράγωνον συνέσταται τὸ ἀπὸ τῆς *EΘ* ἀναγραφησόμενον· ὅπερ ἔδει ποιῆσαι.

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1. *τό*] (tert.) supra m. 1 V. 2. *EH*] *HE* P. 3. *ἴσον* — 5. *HΘ*] mg. m. 2 V; in textu ras. tertiae partis lineae. *ἐστίν* φ. 4. *ὑπὸ τῶν BE, EZ*] *ὑπὸ τῶν BE, EZ ὀρθογώνιον* in mg. transiens m. 1 F, seq. *τῶν BE, EZ* φ; *τῶν BE, EZ περιεχόμενον ὀρθογώνιον* p. 5. *HE*] *HE τετραγώνου* p; *τετραγώνου* add. comp. m. 1 F. *δὲ ἀπό*] euan. F. 6. *ἐστίν* Vφ. *EH*] Pp; *HE* BF, in ras. V. 7. *EZ περιεχόμενον ὀρθογώνιον* p. *HE*] PB; *τῆς HE* Vφ, *τῆς EH* p. 8. *ἴσα*] *ἴσον* φ. *ἐστίν* V. *τοῖς*] in ras. V. *ΘE, EH*] Pp; *ΘE, HE* BF, V in ras. 9. *HE*] *EH* p. *τῶν*] supra m. 2 V. 10. *περιεχόμενον ὀρθογώνιον*] om. p. *ἐστίν* V. *τῷ*] *τό* φ. 11. *τὸ BΔ*] BFVp, Campanus; *τὸ ὑπὸ τῶν BE, EΔ* P. 12. *EZ*] *ZE* P. 13. *ἐστίν* V. 14. *καί*] postea add. comp. F; om. V. *A*] insert. m. 1 p. 15. *ἐστίν* PV. *ἀναγραφησομένῳ*] PBF; *ἀναγραφομένῳ* V, *ἀναγραφέντι* p. 18. *συνέσταται*] BF; *συνίσταται* Pp et V in ras. *ἀναγραφέν*

est in $H$ in inaequales autem in $E$, erunt

$$BE \times EZ + EH^2 = HZ^2 \text{ [prop. V]}.$$

sed $HZ = H\Theta$. itaque $BE \times EZ + HE^2 = H\Theta^2$. uerum $\Theta E^2 + EH^2 = H\Theta^2$ [I, 47]. itaque

$$BE \times EZ + HE^2 = \Theta E^2 + EH^2.$$

subtrahatur, quod commune est, $HE^2$. itaque

$$BE \times EZ = E\Theta^2.$$

uerum $BE \times EZ = B\Delta$; nam $EZ = E\Delta$. itaque $B\Delta = \Theta E^2$. sed $B\Delta = A$. itaque etiam figura rectilinea $A$ quadrato, quod in $E\Theta$ construi poterit, aequale est.

Ergo datae figurae rectilineae $A$ aequale quadratum constructum est, id quod in $E\Theta$ describi poterit; quod oportebat fieri.

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p. 19. *ποιῆσαι*] *δεῖξαι* F V. *Εὐκλείδου στοιχ. β* B, *Εὐκλείδου στοιχείων τῆς Θέωνος ἐκδόσεως β* F, *τέλος τοῦ δευτέρου στοιχείου τοῦ Εὐκλείδου τοῦ γεωμέτρου* V.



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ἔστι δὲ καὶ τὰ ΘΖ, ΓΚ τετράγωνα ἀπὸ τῶν ΑΓ, ΓΒ· τὰ ἄρα τέσσαρα τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ ἴσα ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογωνίῳ. ἀλλὰ τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ ὅλον ἐστὶ τὸ ΑΔΕΒ, ὅ ἐστιν ἀπὸ τῆς ΑΒ τετράγωνον· τὸ ἄρα ἀπὸ τῆς ΑΒ τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογωνίῳ.

Ἐὰν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὡς ἔτυχεν, τὸ ἀπὸ τῆς ὅλης τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν τμημάτων τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογωνίῳ· ὅπερ ἔδει δεῖξαι.

[Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐν τοῖς τετραγώνοις χωρίοις τὰ περὶ τὴν διάμετρον παραλληλόγραμμα τετράγωνά ἐστιν].

ε'.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταξὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ.

Εὐθεῖα γάρ τις ἡ ΑΒ τετμήσθω εἰς μὲν ἴσα κατὰ

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IV. πόρ. De Proclo p. 304 u. ad IV, 15. V. Boetius p. 385, 17.

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1. ἔστιν P. τά] τό F; corr. m. 2. τετράγωνον F; corr. m. 2. 2. τά] (alt.) om. F. ἐστίν P. 3. τε] m. 2 V. 4. ὀρθογώνια φ. τά] τὰ τέσσαρα P. ΘΖ] Θ in ras. V; ΖΘ B. 5. ΗΕ] Η e corr. p. ἐστίν P. ΑΔΕΒ

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(nam $H\Gamma = \Gamma B$), erit etiam $HE = A\Gamma \times \Gamma B$. itaque $AH + HE = 2\, A\Gamma \times \Gamma B$. uerum etiam quadrata $\Theta Z$, $\Gamma K$ in $A\Gamma$, $\Gamma B$ constructa sunt. ergo $\Theta Z + \Gamma K + AH + HE = A\Gamma^2 + \Gamma B^2 + 2\, A\Gamma \times \Gamma B$. sed $\Theta Z + \Gamma K + AH + HE = A\Delta EB = AB^2$. itaque $AB^2 = A\Gamma^2 + \Gamma B^2 + 2\, A\Gamma \times \Gamma B$.

Ergo si recta linea utcunque secatur, quadratum totius aequale est quadratis partium et duplo rectangulo partibus comprehenso; quod erat demonstrandum.[1])

## V.

Si recta linea in partes aequales et inaequales secatur, rectangulum inaequalibus partibus totius comprehensum cum quadrato rectae inter sectiones positae aequale est quadrato dimidiae.[2])

nam recta quaelibet $AB$ in aequales partes sece-

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1) Etiam Campanus hic duas demonstrationes habet, quarum prior reiectae, altera neque huic neque reiectae similis est. de hac habet: „sed hac uia non patet correlarium, sicut uia praecedenti patet, unde prima est autori magis consona.“ nam corollarium et ipse habet. itaque fortasse Theone antiquius est.

2) $ab + \left(\frac{a+b}{2} - b\right)^2 = \left(\frac{a+b}{2}\right)^2$.

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τετράγωνον V. 6. $AB$ τετράγωνον] (prius) mg. m. 2 V; in textu ras. 2—3 litt. τετράγωνον] mg. m. 2 F. 7. ἐστίν P. τε] om. p. τῶν] m. 2 F. 9. ἔτυχεν] B; ἔτυχε uulgo. 10. ἐστίν P. τε] om. p. 12. Sequitur alia demonstratio, quam Augustum secutus in appendicem reieci. 13. πόρισμα — 16. ἐστιν] add. Theon? (BFVp); mg. m. rec. P. 14. τούτων P. φανερόν ἐστιν V. 18. εἰς] supra m. 1 V. 19. εἰς ἄνισα p. 21. ἐστίν P.



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δὲ ἡ ὑπὸ $BA\Delta$· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $A\Delta E$. τῶν δὲ παραλληλογράμμων χωρίων αἱ ἀπεναντίον πλευραί τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν· ὀρθὴ ἄρα καὶ ἑκατέρα τῶν ἀπεναντίον τῶν ὑπὸ $ABE$, $BE\Delta$ γωνιῶν· ὀρθογώνιον ἄρα ἐστὶ τὸ $A\Delta EB$. ἐδείχθη δὲ καὶ ἰσόπλευρον.

Τετράγωνον ἄρα ἐστίν· καί ἐστιν ἀπὸ τῆς $AB$ εὐθείας ἀναγεγραμμένον· ὅπερ ἔδει ποιῆσαι.

## μξ'.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτεινούσης πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιεχουσῶν πλευρῶν τετραγώνοις.

Ἔστω τρίγωνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν ἔχον τὴν ὑπὸ $BA\Gamma$ γωνίαν· λέγω, ὅτι τὸ ἀπὸ τῆς $B\Gamma$ τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν $BA$, $A\Gamma$ τετραγώνοις.

Ἀναγεγράφθω γὰρ ἀπὸ μὲν τῆς $B\Gamma$ τετράγωνον τὸ $B\Delta E\Gamma$, ἀπὸ δὲ τῶν $BA$, $A\Gamma$ τὰ $HB$, $\Theta\Gamma$, καὶ διὰ τοῦ $A$ ὁποτέρᾳ τῶν $B\Delta$, $\Gamma E$ παράλληλος ἤχθω ἡ $A\Delta$· καὶ ἐπεζεύχθωσαν αἱ $A\Delta$, $Z\Gamma$. καὶ ἐπεὶ ὀρθή ἐστιν ἑκατέρα τῶν ὑπὸ $BA\Gamma$, $BAH$ γωνιῶν, πρὸς δή τινι εὐθείᾳ τῇ $BA$ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ $A$ δύο εὐθεῖαι αἱ $A\Gamma$, $AH$ μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεξῆς γωνίας δυσὶν ὀρθαῖς ἴσας ποιοῦσιν· ἐπ' εὐθείας ἄρα ἐστὶν ἡ $\Gamma A$ τῇ $AH$. διὰ τὰ αὐτὰ δὴ καὶ

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XLVII. Pappus I p. 178, 11. Schol. in Archim. III p. 383. Boetius p. 384, 21.

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1. *καί*] insert. m. rec. b (comp.). 5. *ἐστίν* PV; comp. b.

[prop. XXIX]. uerum $\angle BA\Delta$ rectus est. itaque etiam $\angle A\Delta E$ rectus. sed in spatiis parallelogrammis latera angulique opposita inter se aequalia sunt [prop. XXXIV]. itaque etiam uterque angulus oppositus $ABE$, $BE\Delta$ rectus est. rectangulum igitur est $A\Delta EB$. demonstratum autem est, idem aequilaterum esse. ergo quadratum est [def. 22]. et in recta $AB$ constructum est; quod oportebat fieri.

## XLVII.

In triangulis rectangulis quadratum in latere sub recto angulo subtendenti constructum aequale est quadratis in lateribus rectum angulum comprehendentibus constructis.

Sit triangulus rectangulus $AB\Gamma$ rectum habens $\angle BA\Gamma$. dico, esse $B\Gamma^2 = BA^2 + A\Gamma^2$.

construatur enim in $B\Gamma$ quadratum $B\Delta E\Gamma$, in $BA$, $A\Gamma$ uero $HB$, $\Theta\Gamma$ [prop. XLVI], et per $A$ utrique $B\Delta$, $\Gamma E$ parallela ducatur $A\Delta$ [prop. XXXI]; et ducantur $A\Delta$, $Z\Gamma$. et quoniam rectus est uterque angulus $BA\Gamma$, $BAH$, ad rectam quandam $BA$ et punctum in ea situm $A$ duae rectae $A\Gamma$, $AH$ non in eadem parte positae angulos deinceps positos duobus rectis aequales efficiunt; itaque in eadem recta sunt $\Gamma A$, $AH$ [prop. XIV]. eadem igitur de causa etiam

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τὸ $A\Delta EB$] mg. m. 2 V; in F supra $E$ scr. $H$. 7. ἐστίν] (prius) PF; ἐστί uulgo. 12. τὴν] περὶ τὴν Proclus. 13. περιεχουσῶν] om. Proclus. 15. $BA\Gamma$] corr. ex $B\Gamma A$ m. 2 F. Ante $B\Gamma$ eras. $A$ P. 16. ἴσον] supra m. 2 (comp.) F. ἐστίν P. $BA$] $AB$ F. 18. μέν] om. F. 19. $B\Gamma\Delta E$ F. $HB$] corr. ex $BH$ m. 2 F. $\Theta\Gamma$] $\Gamma$ in ras. est in F; seq. in V m. 2: τετράγωνα. 20. ἤχθω παράλληλος p. $A\Delta$] $\Delta$ in ras. P m. 1. 23. $BA$] $AB$ p. 26. τὰ αὐτὰ] ταῦτα Bp.

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